

VIDIT RAJESH PRABHU

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EDUCATION

Master of Science in Computer Science and Engineering <i>University at Buffalo</i>	2024 – 2026 GPA: 3.6 / 4.0
Bachelor of Engineering in Electronics and Telecommunication Engineering <i>Don Bosco Institute of Technology, Mumbai</i>	2018 – 2022 GPA: 3.6 / 4.0

EXPERIENCE

ML/AI Engineering Intern <i>FourSteps Solutions</i>	Apr 2026 – Present <i>Fremont, CA</i>
- Resolved a production security vulnerability in a FastAPI backend by redesigning the RBAC permission architecture to enforce least-privilege access across roles, closing a billing data exposure risk - Proposed and shipped a new permission layer during code review that separated internal analytics from customer-facing access, preventing a potential data leak in a live PostgreSQL-backed system - Built a pytest suite verifying the full permission matrix against live SQLAlchemy models and independently traced end-to-end data flow across a large multi-repo Python codebase	
Software Development Intern <i>Bluebird AI</i>	Jun 2025 – Aug 2025 <i>San Ramon, CA</i>
- Delivered a voice assistant from scratch as sole backend engineer integrating Google Speech-to-Text and Gemini LLM covering voice and text input, intelligent LLM routing, live database retrieval, and plain-language response generation - Cut average response latency by approximately 30% measured via log timing by rearchitecting API concurrency, restructuring inference execution, and deploying behind Nginx to hold up under real load - Drove out-of-context LLM response rate from roughly 1 in 6 queries to near zero by pairing structured query routing with targeted prompt engineering ahead of user-facing launch	
Assistant System Engineer <i>Tata Consultancy Services Ltd.</i>	Jun 2022 – May 2024 <i>Mumbai, India</i>
- Contributed to a 15% team-wide gain in order processing efficiency by tuning SQL queries and profiling bottlenecks across a Siebel CRM serving 3,000+ users - Kept a high-traffic CRM stable across releases by owning backend workflow fixes, UI changes, and field-level modifications through fast moving Agile sprints	

PROJECTS

AlphaSignal – Financial News Sentiment & Market Intelligence Agent
- Achieved 88.3% accuracy in financial news sentiment classification by fine-tuning FinBERT on 11,931 labeled samples, with full experiment tracking across multiple runs via MLflow - Built a LangGraph agentic pipeline with RAG-powered retrieval over live market news, Redis caching for sub-100ms repeated queries, and automated deployment via GitHub Actions CI/CD
Clairvoyix – Private GPU Powered AI Compute Platform
Engineered a private GPU-powered ML orchestration platform for the AEC industry, containerizing BART, SAM2, and a T5 RAG pipeline across Docker microservices with a Flask backend for job routing, reducing manual processing effort by 40% with cost estimates within a 10% margin
Distributed In-Memory Cache System
Built a distributed in-memory cache from scratch in Python with consistent hashing across 3 nodes, write replication, and automatic heartbeat failover sustaining 4137 req/s throughput at 100% cache hit rate with 10.9ms median GET latency and sub-53ms p99 across 10000 concurrent requests

SKILLS

Languages: Python, C, C++, Java, SQL, JavaScript
Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, Keras, OpenCV, FastAPI, Flask, LangGraph, LlamaIndex, React
Tools & Infrastructure: AWS, Docker, Redis, PostgreSQL, Git, CI/CD, MLflow, ChromaDB, Prometheus, LLMs